FROM THE
EDITOR-IN-CHIEF

Climate chaos appears to be the new normal for India. We are inundated by visual evidence of the havoc it's causing. Much of Punjab resembles a sea: some 1,900 villages across 23 districts under water, after a record rainfall of 1,000 per cent above normal in the first week of September. The toll is stark: 56 dead, nearly half a million acres of farmland submerged, about Rs 12,000 crore lost. The images from neighbouring Himachal Pradesh, Uttarakhand, Jammu and Kashmir are equally frightening: monster floods washing away villages, hillsides crumbling and turning bridges, lorries and riverside hotels into floating debris. Himachal has counted 45 cloudbursts, 91 flash floods and 105 major landslides just in this one season. It's a cautionary tale. For, India is now among the top ten nations most vulnerable to the impact of climate change.

In fact, extreme weather events, be it floods, droughts, cyclones or heat waves, are devastating India with increasing intensity and frequency every year. In 2022, India saw such events on 314 days. That rose to 318 days in 2023 and to 322 days by 2024. That's an alarming 88 per cent of 366 days last year that Indians were in a weather emergency. This year, in the 90 days from January to March, India saw extreme weather events on 87 days. Together, they exacted a brutal toll: 3,080 climate-related deaths in 2024–25, the highest in a decade. A June report by development consultancy IPE Global and advanced mapping solutions provider Esri India found 85 per cent of India's 738 districts to be vulnerable to extreme events, and the costs are mounting. For 1998-2017, a UN report pegged India's total losses in climate disasters at $79.5 billion (Rs 7 lakh crore). By 2030, the RBI estimates that up to 4.5 per cent of India's GDP (Rs 11.5 lakh crore of the projected Rs 255 lakh crore GDP at constant prices) may be at risk from labour hours lost to extreme heat and humidity.

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For India, global warming is no more a myth. The 10 warmest years since 1900 have all occurred in the last 15 years. Average temperatures here rose 0.69°C in just the past two decades. That may look modest, but in climate science each minute degree has an exponential effect. In fact, the entire planet is heating up at breakneck speed. The red lines have been crossed. The World Meteorological Organization declared 2024, the warmest in 175 years of record-keeping, as the first year that average global temperatures breached the 1.5°C threshold above pre-industrial levels. The 2015 Paris Climate Accord accepts that exceeding 2°C will take global conditions to levels never experienced before by human civilisation. Put simply, global warming is a result of massive amounts of carbon emissions caused by burning coal, oil and gas besides rampant deforestation in the past century. This causes oceans and water bodies to evaporate faster, leading to major disturbances in global climate patterns. For every one degree rise in global temperatures, moisture in the atmosphere is estimated to rise by 7 per cent. This then pours down in huge quantities on vulnerable zones.

India is now a prime victim. The current havoc in the hills and northern plains, which our cover story chronicles, is direct proof. What has become critical is the climate mitigation and adaptation strategies India needs to adopt. Unfortunately, states have lagged on these measures and are paying the price. Punjab was warned about flood risk as early as April but took no mitigating action. Desilting of rivers and dams was not done on priority in Himachal. Instead, it has blithely sanctioned buildings and infrastructure on vulnerable riverbanks and hillsides, as have Uttarakhand and Kashmir. Major cities, too, are victims of civic apathy and corruption, with Delhi, Mumbai, Bengaluru and Chennai regularly going under due to poor drainage management. "The biggest problem is the lack of implementation of environmental bylaws by local bodies, with the connivance of contractors, government agencies and political patronage violating all safety norms," says Prof. Sanjay Srivastava, who chairs an inter-governmental body on climate resilience.

Correctives exist, but they demand urgency. Experts call for strict regulation of construction, roads, drainage and power supply. Cities must be redesigned as "urban sponges"—with permeable pavements, parks, rain gardens and effective drainage to soak up excess water. Dams must be regularly desilted to restore capacity. Encroachments on riverbanks and hillsides must be banned. Disaster management forces must be strengthened, evacuation routes mapped, shelters prepared and awareness spread. Odisha has shown how foresight and rigorous protocols can save lives during cyclones. The same culture of preparedness must spread nationwide.

Equally, we need hyperlocal forecasting so that vulnerability can be plotted down to the kilometre. Adaptation also requires climate-sensitive policies across sectors. Agriculture must rapidly shift to hardier, drought-resilient crops. Urban planning must integrate rainfall and drainage data. And the root cause must be addressed. The government's commitment to net zero carbon emissions by 2070 and 50 per cent energy through renewables by 2030 is the right direction. But ambition must translate into relentless execution.

Climate change is no longer tomorrow's threat; it is today's lived reality. India cannot afford apathy or short-cuts. Prevention, not *jugaad*, must be the national mindset. Only then can we hope to shield our people from a future of permanent emergencies.

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